

Job Seeker Copilot

Engineering a multi-service job-search and application platform

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Software Developer

Controlled private beta

The screenshot displays the Job Seeker Copilot interface, which is divided into three main sections: Search, Applications, and Documents. The Search section is active, showing a list of job matches for the role of Software Developer. The results are sorted by 'Most relevant' and include details such as the job title, company, location, salary, and posting date. The Applications section shows a list of applications with a total of 6. The Documents section shows a list of documents with a total of 6. The interface also includes a sidebar with navigation links and a bottom section for recent activity.

Search

REAL PROVIDERS Job search, matching and details

Job Matches Real providers Sort: Most relevant Filter Refresh

Software Developer (188) Software Engineer (Not searched) Full-Stack Developer (Not searched)

Java Developer (Not searched) Backend Developer (Not searched)

SHOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER · MORE AVAILABLE

Software Engineer REED.CO.UK DOCUMENTS PREPARED

Vermillion Analytics

London GBP 35,000 - 45,000 per year Posted Jun 8, 2026

Lead Full Stack Software Developer REED.CO.UK NOT SAVED

Spectrum IT Recruitment

Guildford GBP 80,000 - 100,000 per year Posted May 8, 2026

Junior Software Engineer REED.CO.UK DOCUMENTS PREPARED

Harnham - Data & Analytics Recruitment

London GBP 30,000 - 40,000 per year Posted Aug 7, 2026

Software Engineer (C/C+) REED.CO.UK DOCUMENTS PREPARED

Rise Technical Recruitment Limited

Stevenage Salary not specified Posted date unavailable

Applications 6

Generated documents 6 100%

Applied 0 0%

Interviews 0 0%

Offers 0 0%

Unsuccessful 0 0%

Recent activity

11/08 20:36 Documents

Downloaded previous cover letter version 1.

NOT ENABLED FOR THIS BETA

AI Credit & Payments

Wallet balance and purchasing will be enabled after the secure payment integration is approved.

Open AI Credit area

AVAILABLE Reporting & job-search evidence

Your job search progress Refresh

29 hours remaining this week.

An evidence-backed product dossier for recruiters, engineering teams and potential partners. Private source access and deeper technical evidence can be provided on request.

Product first. Engineering evidence underneath.

The first pages explain the product in plain language. Later pages show how its contracts, state owners, provider boundaries, recovery controls and tests make it more than a user-interface prototype.

03 **Executive overview**

05 **Product journey**

07 **Backend design**

09 **Engineering around an LLM**

11 **Applications and reporting**

13 **Runnable environments**

15 **Challenges and lessons**

04 **What I built**

06 **System architecture**

08 **Providers and location**

10 **Document evidence**

12 **Reliability and testing**

14 **Integrity, privacy and safety**

16 **Status and professional evidence**

28

healthy application services in the verified
manual runtime

32/32

authoritative regression scenarios passed
in the closure evidence

25

concurrent active fixture sessions
completed; 15 is the cautious operating
point

Evidence boundary. Screenshots are privacy-reviewed crops from the working local product. Live-provider labels mean real external responses; they do not imply provider uptime, public production deployment or measured employment outcomes.

One workspace for the fragmented work of finding a job

Job Seeker Copilot helps a candidate organise evidence, discover vacancies, prepare grounded application documents, track progress and review activity without repeatedly rebuilding the same context.

The product goal is not “AI writes a CV”. It is a traceable workflow from verified career evidence and a real vacancy to an application-specific document set.

The current software is a **controlled private beta**. Its normal local environment can use live Reed, Adzuna, JSearch, NHS Jobs, Find an apprenticeship, Postcodes.io, Google Maps and OpenAI paths. The default engineering environment remains deterministic, and live Stripe checkout is not enabled.

Bernard conceived the product, set its architecture and product direction, coordinated a multi-repository implementation, integrated the service boundaries, ran realistic acceptance work and repeatedly debugged failures exposed by real-user journeys.

Problem

Job search information is scattered across boards, CV files, notes and application trackers. Generic AI output can introduce unsupported claims.

Product response

Structured candidate evidence, provider-normalised vacancies, durable workflows, exact document versions and application tracking in one authenticated interface.

Engineering response

Domain-aligned services, producer-owned contracts, deterministic provider modes, fail-closed validation and recovery designed around unreliable external systems.

31

workspace repositories audited: runtime/test catalogue, infrastructure and landing

7

independent PostgreSQL state owners

5

current job-provider integrations

4

document outputs per approved CV/letter pair: DOCX + PDF

Full-stack delivery, integration and product ownership

Candidate experience

- ✓ Angular/TypeScript authenticated workspace
- ✓ Progressive profile and reusable Evidence Library
- ✓ Live/fixture job discovery with explicit source state
- ✓ Applications, documents, version history and reporting
- ✓ Accessible states for loading, absence and provider failure

Java/Spring platform

- ✓ Gateway and domain-service boundaries
- ✓ REST/OpenAPI contracts and generated clients
- ✓ PostgreSQL/Flyway state ownership
- ✓ Idempotent multi-service workflows
- ✓ External provider mapping, timeouts and degradation

AI document engineering

- ✓ Purpose-bound evidence snapshots
- ✓ Versioned prompts and strict output schemas
- ✓ Claim ledger, structural repair and grounding
- ✓ Unknown-outcome reconciliation and bounded retry
- ✓ Deterministic evidence-based CV fallback

Delivery system

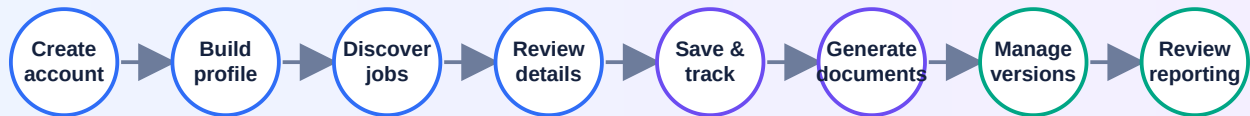
- ✓ Docker Compose profiles and persistent volumes
- ✓ GitHub branch/PR/CI delivery across repositories
- ✓ Playwright/Cucumber E2E and promo automation
- ✓ Capacity, cost and AWS decision evidence
- ✓ Central documentation and reproducible contracts

The professional evidence is the ability to design, direct, integrate, test, debug and evolve a substantial product—not a claim that every line was typed without modern engineering assistance.

How the work was approached

Work was broken into explicit features and service-owned changes, reviewed through source, contracts and running behaviour, then reconciled through integration branches. Realistic manual use exposed defects that synthetic journeys missed; those failures were treated as evidence to improve the design rather than hidden from the narrative.

From a reusable profile to a traceable application



Discover. A real-provider search returned 188 current results for the selected role. Provider source and saved/document state remain visible.

The screenshot shows a search interface with filters for 'Software Developer' (188 results), 'Software Engineer' (Not searched), 'Full-Stack Developer' (Not searched), 'Java Developer' (Not searched), and 'Backend Developer' (Not searched). Below the filters, it says 'SHOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER - MORE AVAILABLE'. The results list includes:

- Software Engineer** (Vermillion Analytics, London, GBP 35,000 - 45,000 per year, Posted Jun 8, 2026) - REED.CO.UK DOCUMENTS PREPARED
- Lead Full Stack Software Developer** (Spectrum IT Recruitment, Guildford, GBP 80,000 - 100,000 per year, Posted May 8, 2026) - REED.CO.UK NOT SAVED
- Junior Software Engineer** (Harnham - Data & Analytics Recruitment, London, GBP 30,000 - 40,000 per year, Posted Aug 7, 2026) - REED.CO.UK DOCUMENTS PREPARED
- Software Engineer (C/C+)** (Rise Technical Recruitment Limited, Stevenage, Salary not specified, Posted date unavailable) - REED.CO.UK DOCUMENTS PREPARED
- Software Developer Trainee** (ITOL Recruit, E62RP, GBP 28,000 - 35,000 per year, Posted date unavailable) - REED.CO.UK NOT SAVED
- Lead AI Software Engineer** (Avanti, W60QU, GBP 80,000 - 120,000 per year, Posted date unavailable) - REED.CO.UK NOT SAVED

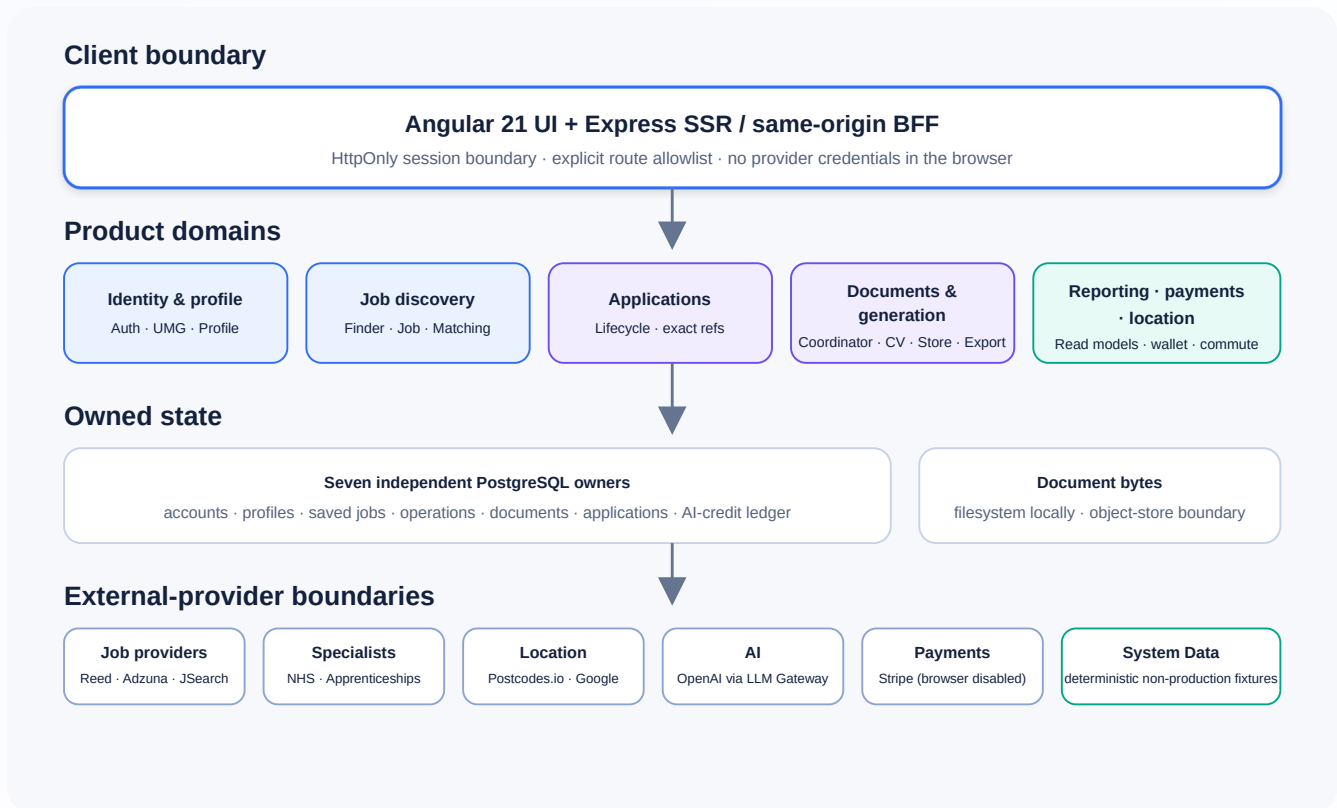
Track. Applications preserve vacancy identity, lifecycle milestones, evidence choices and the exact prepared document state.

The screenshot shows the application details for 'Software Engineer' (Harnham - Data & Analytics Recruitment, London). It includes a timeline with milestones: CV Generated (11 Aug, 20:34), Cover Letter Generated (11 Aug, 20:34), Documents Uploaded, Mark Applied, Interview, and Offer. Below the timeline, there are sections for 'Documents for this application' (Choose exact approved versions; Recommended does not save automatically), 'Experience & achievements used' (Selected for the current document drafts), and a 'View Application' button. The application is marked as 'DOCUMENTS PREPARED'.

What makes the journey connected

Search does not create a hidden application. A job is explicitly saved, generation uses the saved immutable snapshot, the user approves exact drafts, and Application Tracker records exact document references. Document history can evolve without silently changing what an application used.

Decomposed by ownership, not by repository count



Why gateways?

Browser gateways constrain routes and identity. Provider gateways isolate external credentials, DTOs and failure translation.

Why separate state?

Each persistent domain owns its PostgreSQL schema. Cross-service workflows use operation records and idempotency rather than shared tables.

Why fixtures?

The same boundaries can run with governed System Data responses, keeping CI repeatable and preventing accidental spend.

Representative services and their decisions

Job Service

Bounds provider fan-out, maps adverts into a canonical model, retains field/source provenance, deduplicates related adverts and persists only explicit saved-job snapshots.

User Profile Service

Owns subject-scoped career data, search preferences, revisioned evidence and purpose-bound immutable snapshots used by generation.

Document Generation Gateway

Coordinates saved-job resolution, evidence snapshots, AI-credit reservation, drafts, approval, export and Tracker linkage through a durable PostgreSQL operation ledger.

Document Store

Owns families, immutable versions, current/approved pointers, lineage, file metadata and bytes. Local uploads cross an isolated ClamAV scan boundary.

Application Tracker

Owns lifecycle and immutable events. It independently verifies document ownership and commits exact version references atomically.

Reporting

Builds read-only summaries, recent activity, an indicative time estimate and an evidence export from owner-scoped upstream data.

Contract discipline



Repositories build without copied local client JARs. Consumers pin reviewed producer revisions; generated output remains disposable build material.

Deliberate honesty: "Job Matching" currently means reconciliation with existing application state plus advisory commute metadata. It does not calculate a candidate-skill suitability score.

Live boundaries are explicit—and optional

LIVE / REAL PROVIDER

FIXTURE / DETERMINISTIC

DISABLED / NOT EXPOSED

Boundary	Purpose	Current controlled-beta state
Reed, Adzuna, JSearch	General vacancy discovery	● Bounded live samples; fixture default
NHS Jobs	Official health-sector vacancies	● Live XML path; deterministic in-service fixture
Find an apprenticeship	DfE vacancies and training detail	● Authenticated live snapshot; fixture default
Postcodes.io	Persistable UK postcode/place authority	● Public live API supported; fixture default
Google Places / Routes	Autocomplete and advisory commute	● Explicit live profile; five-result browser proof
OpenAI	Typed CV/cover-letter generation	● Explicit paid profile; deterministic default
Stripe	Checkout and signed webhook	● Backend fixture; browser checkout disabled

Location and commute [Edit](#)

Reading · within miles

Town or postcode

Reading

5 matching locations found.
Suggestions provided by Google Maps

- Reading
- Reading Street, Tenterden
- Whitley, Reading
- Reading Street, Broadstairs
- Newtown, Reading

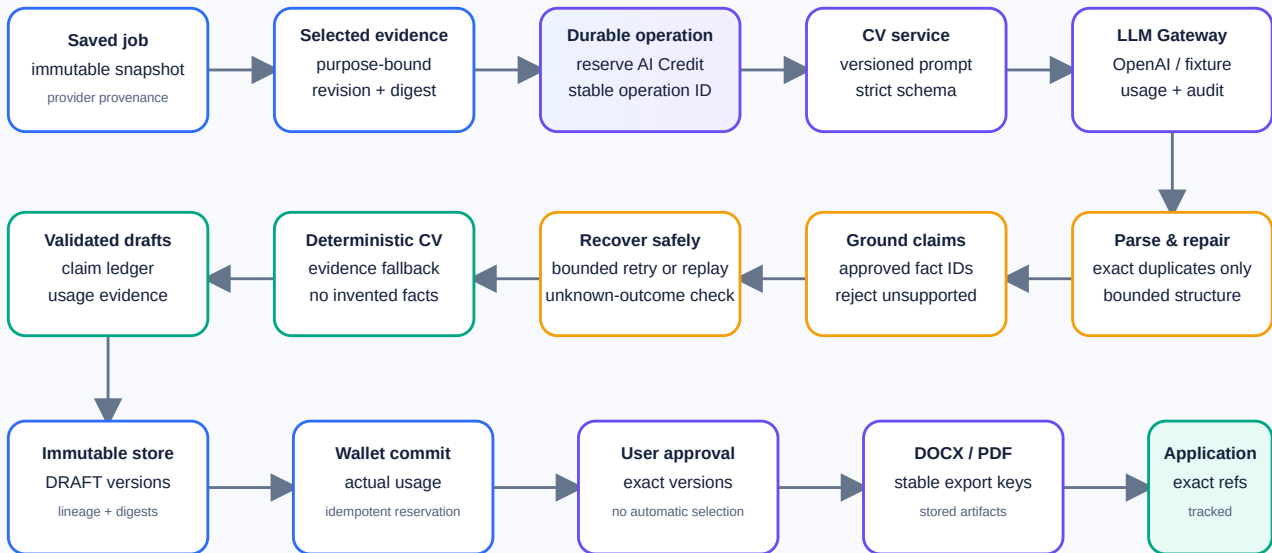
A normal UI request returned five suggestions with required Google Maps attribution.
The search text was not saved.

Location ownership

- ✓ Browser calls the same-origin BFF.
- ✓ Location Gateway owns the public contract.
- ✓ Location Service owns provider selection and canonical rules.
- ✓ Google Maps Gateway alone receives the key and provider DTOs.
- ✓ Postcodes.io provides persistable UK canonical fields.
- ✓ Route estimates remain transient and advisory.

Engineering around an unreliable model

The LLM is a bounded collaborator, not a source of truth



Untrusted output

The model response is parsed against a strict schema, bounded, repaired only for harmless structure, and checked again against approved fact IDs.

Safe recovery

Retry is bounded to classified safe failures. Unknown outcomes are reconciled before another paid call. Retained responses can be replayed without calling the model again.

Fail useful

When acceptable model CV content cannot be obtained, an evidence-based deterministic CV can complete the workflow without invented qualifications or metrics.

Generation is a stateful product operation

Software Engineer
Vermillion Analytics
London GBP 35,000 - 45,000 per year Posted Jun 8, 2026

Job Description
Software Engineer Location: Hybrid / Remote (UK-based) Team: Engineering Salary: Up to 45000 Reports to: VP of Engineering You must be UK based and eligible to work. About Our Client Our client is one of Europe's fastest-growing software companies, on a mission to solve productivity problems for people-intensive industries using AI and automation. Since being founded, they've focused on automating the manual, tedious parts of complex workflows...

[Read full advert](#)

[View on Reed.co.uk](#)

✓ **CV and Cover Letter generated successfully**

CV
Download DOCX Download PDF Upload CV

COVER LETTER
Download DOCX Download PDF Upload Letter

Real vacancy, real provider path and successful CV/cover-letter operation. Four retained download choices are presented only after the exact drafts complete.

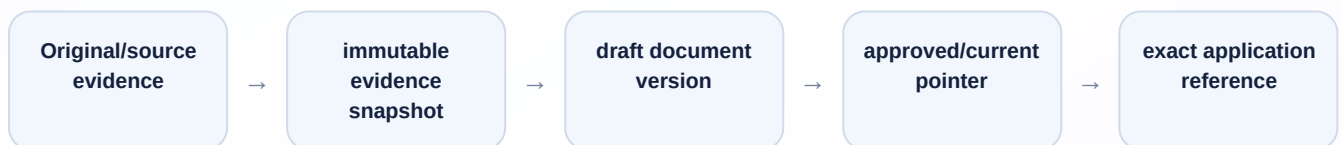
Demonstrated here

- ✓ Saved provider vacancy
- ✓ Selected candidate evidence
- ✓ CV and cover letter prepared
- ✓ DOCX and PDF available
- ✓ Application state updated

Not silently claimed

A green result does not prove every future model call will succeed. The engineering claim is that failures are bounded, diagnosable and recoverable without inventing content or double-charging the wallet.

Version and lineage model



Document families group history; versions remain immutable. Changing the family's current version does not rewrite a submitted application's frozen selection.

17

bounded live OpenAI calls in the closure
quality/cost sample

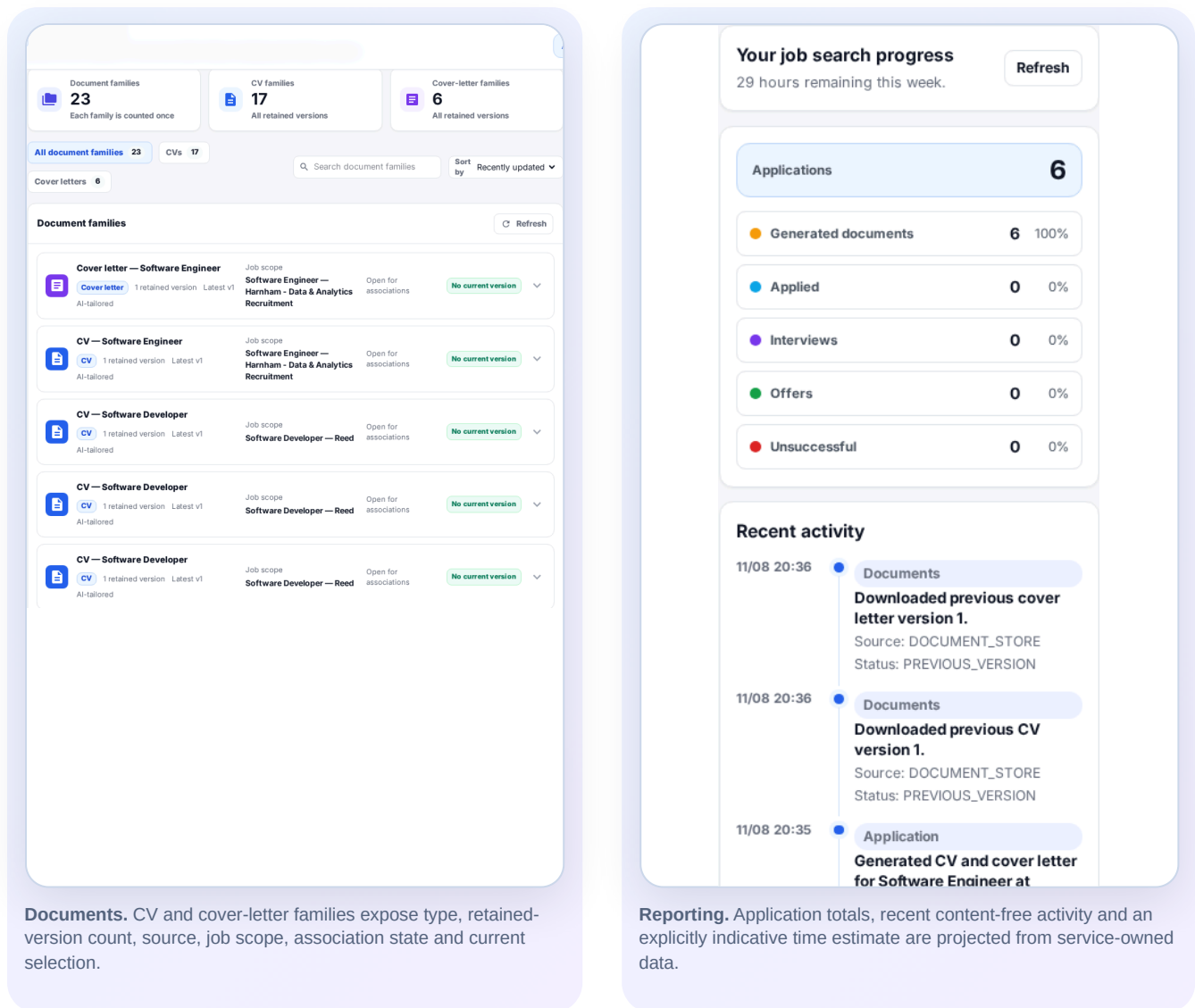
\$0.069

observed OpenAI cost for that sample—
not a usage forecast

0

live Stripe charges in the validated
environment

State remains visible after generation



Application truth

Tracker owns status, immutable events and exact document references.

Document truth

Store owns families, versions, files, lifecycle and lineage.

Reporting truth

Reporting reads owner-scoped APIs and never decides or mutates domain state.

Product discipline: a “recommended” document does not save itself. The candidate selects or explicitly omits the exact version for each application slot.

Confidence is layered—and bounded

Fast feedback

Unit/component tests, parser/policy tests, frontend specs, migration tests and generated-client checks.

Boundary confidence

Producer contracts, semantic compatibility, HTTP integration, PostgreSQL persistence and container verification.

Product confidence

Cucumber/Playwright journeys, accessibility automation, real-provider samples, manual acceptance and capacity workloads.

32

regression scenarios

244

passed journey steps

7

canonical personas covered

26

steps in the connected promo journey

Realistic testing changed the product

1

Document/application association.

A completed generation was not enough; exact retained versions also had to become visible and selectable on the correct application.

2

LLM duplicate and timeout behaviour.

Fail-closed validation evolved into narrow structural repair, bounded retry, outcome reconciliation, quarantine replay and fallback.

3

Rich real profiles.

Real career structure exposed selection and formatting assumptions that compact deterministic profiles did not.

4

Provider detail inconsistency.

Canonicalisation and source attribution matter because provider IDs, dates, descriptions and locations vary.

What the numbers do not prove: universal provider uptime, a complete manual accessibility audit, public production security, employment outcomes or 50-user capacity.

Runnable software, not a collection of repositories

Supported Compose profiles

basic-fixture	Account, profile, location and job search
full-fixture	Complete deterministic application
full-local-ses	Fixture app + production email adapter against LocalStack
google-maps-smoke	Only Google live; other providers fixture-backed
real-job-providers	Five live job sources + Postcodes.io; fixture LLM/Stripe
real-providers	Live jobs, Postcodes.io, Google and OpenAI; fixture Stripe

Current evidence

The supported `real-providers` lifecycle reported 28 healthy application services. Persistent PostgreSQL and document volumes survived ordinary restarts; personal data and credentials remain outside Git.

Reproducibility controls

- ✓ Single service/dependency/port catalogue
- ✓ Reviewed workspace revision lock
- ✓ Generated owner-only runtime environments
- ✓ Explicit live-provider overlays
- ✓ Health and mode verification before use
- ✓ Workspace-local build caches

Capacity evidence

Fixture-backed 1/5/10/15/20/25 active-browser ladders completed. At 25, p95 reached 59.45 seconds, so the documented cautious local operating point is 15—not a public capacity promise.

AWS status

Deployment shapes and costs were modelled from local resource evidence, but the authenticated application has **not** been load-tested or deployed on AWS. The separate landing/waitlist stack uses AWS services and has its own launch gates.

Controls follow the data—not a compliance slogan

Trusted identity

Browser tokens remain in HttpOnly cookies. BFF route allowlists and CSRF checks constrain browser actions; service owners derive identity from validated subjects and dedicated service tokens.

Immutable lineage

Evidence revisions, saved-job snapshots, claims, document versions and application selections carry stable IDs/digests so later edits do not rewrite history.

Secret and personal-data boundary

Provider credentials are runtime-only. Real candidate data, CVs, cookies and passwords are excluded from fixtures, logs, screenshots and Git history.

Untrusted content

Provider text renders as text, external links pass an allowlist, uploads are bounded and scanned, and model output must satisfy schema and evidence policy.

Wallet safety

Reservations, stable operation keys, idempotent commit/release and unknown-outcome checks prevent an ordinary retry becoming a second charge.

Provider minimisation

Google DTOs and keys stay in one gateway; location labels/routes remain transient and attributed. Postcodes.io supplies persistable UK fields.

Scope boundary. These are implemented design controls, not a claim of certification, complete penetration testing, formal regulatory compliance or production-grade operational security.

Shareable-asset review

- ✓ No API keys, tokens, cookies, passwords or private environment values.
- ✓ No real CV, personal email address or phone number embedded in this dossier.
- ✓ Application screenshots exclude account identity/contact information.
- ✓ Private repositories are described, not linked as if publicly accessible.

Failures became architecture

1

LLM output could be fluent and still structurally or factually wrong.

The response: strict typed output, a claim ledger, exact evidence paths, narrow repair, fail-closed grounding, retained-response replay and deterministic CV fallback.

2

Multiple job sources describe the same opportunity differently.

The response: provider adapters, canonical source references, field provenance, URL/content deduplication and partial-provider status instead of a false all-or-nothing success.

3

A generated file is not the same as an application decision.

The response: immutable families/versions, explicit approval/current pointers and exact application selections owned by separate services.

4

Independent repositories create contract drift risk.

The response: producer-owned OpenAPI, reviewed revision/checksum pins, reproducible generated clients and semantic compatibility gates.

5

Deterministic tests overstated confidence in realistic use.

The response: retain canonical personas for repeatability, then add rich-profile, real-provider, paid-LLM and restart-persistence checkpoints with explicit cost authority.

6

Persistent manual state and fixture state have different needs.

The response: named fixture environments for engineering and a separately protected persistent real-user environment that survives ordinary restarts.

Mature engineering confidence is not “nothing failed”. It is knowing what failed, why the previous tests missed it, and what design now makes the failure safer.

What is working—and what this project demonstrates

Working / demonstrated

- ✓ Authenticated profiles and revisioned evidence
- ✓ Five-source fixture and bounded live job discovery
- ✓ Canonicalisation, dedupe and saved snapshots
- ✓ Applications, document lineage and reporting
- ✓ Grounded CV/cover-letter generation and export
- ✓ Live Google autocomplete; advisory commute path
- ✓ Persistent 28-service local manual environment

Known limits / post-beta

- No skill-based suitability score.
- Live Stripe/browser purchasing remains disabled.
- Main application is not deployed or load-tested on AWS.
- Provider/LLM samples do not predict uptime or quality.
- Accessibility automation is not a complete manual audit.
- No measured user, revenue or employment-outcome claim.

Technical stack evidenced by source



Professional capabilities demonstrated

Full-stack delivery · Java/Spring engineering · Angular/TypeScript · API and data design · microservice integration · debugging · testing · AI safety/recovery · Docker · Git/GitHub workflows · technical documentation · product ownership

Job-specific CV and cover letter explain suitability for a role. This portfolio is the engineering evidence behind those claims.

Contact details are provided in the accompanying CV. Demonstration videos include a 204.4-second master showcase and seven feature clips. Private repository access can be arranged on request.